Receding Horizon Control for Safe and Ergonomic Human-Drone Rendezvous with Free-Final-Time

Abstract: This paper addresses the problem of human-drone rendezvous in indoor industrial environments, where safe, ergonomic, and efficient interaction with a moving operator is required. A continuous-time receding horizon control framework with free-final-time is proposed, where the maneuver duration is treated as an optimization variable and updated online. The formulation enables joint optimization of trajectory and completion time under time-varying terminal conditions induced by human motion. Safety is enforced through a distance-dependent speed profile, while ergonomic feasibility is incorporated as a spatial constraint on the rendezvous configuration consistent with the Rapid Upper Limb Assessment method. Efficiency is achieved by minimizing control effort while ensuring smooth and dynamically feasible trajectories. Simulation results in a realistic warehouse scenario demonstrate that the proposed method enables safe and ergonomic interaction with a moving operator, achieving high terminal accuracy under time-varying conditions.

Keywords: Autonomous rendezvous, human-drone interaction, safety, ergonomics, efficiency, receding horizon with free-final-time

1. INTRODUCTION

The progressive digitalization of industrial environments and the advent of Industry 4.0 are fostering the integration of autonomous aerial systems within human-centered production processes. Uncrewed Aerial Vehicles (UAVs) are increasingly employed in collaborative logistics for pick-and-delivery, inspection, and transportation operations, enabling improved flexibility and operational efficiency (Wawrla et al. (2019)). In indoor environments, close proximity between aerial robots and human operators introduces significant challenges related to motion coordination, workspace sharing, and preservation of human well-being. Human–Drone Interaction (HDI) extends the principles of Human–Robot Interaction (HRI) to aerial systems, with the objective of ensuring safe, ergonomic, and efficient cooperation in industrial tasks (Proia et al. (2025)).

Recent literature mainly focuses on outdoor or service-oriented HDI applications, such as photography, surveillance, and inspection (Mirri et al. (2019)), while contributions addressing indoor industrial scenarios remain limited (Proia et al. (2025)). In such contexts, trajectory planning must guarantee safety, ergonomic feasibility, and energy efficiency under dynamic and constrained conditions. Safety is typically enforced through collaborative robotics methodologies, including Speed and Separation Monitoring and Power and Force Limiting (Proia et al. (2021); Lucci et al. (2020)), which regulate motion and proximity in shared workspaces. Ergonomic feasibility is addressed by minimizing biomechanical strain and preserving postural comfort, often quantified through metrics such as the Rapid Upper Limb Assessment (RULA) (McAtamney and Corlett (1993)), used to define admissible interaction regions (Proia et al. (2023, 2025)). Energy efficiency further motivates the design of trajectories that reduce control effort while maintaining task performance.

In pick-and-delivery operations, the descent of a drone toward a moving operator represents a critical phase of HDI. The aerial platform must ensure safe proximity and accurate spatial alignment while satisfying dynamic feasibility, ergonomic constraints, and energy efficiency at the delivery configuration. The corresponding control problem can be formalized as an autonomous rendezvous, in which the drone trajectory is required to converge to a terminal state consistent with the human motion.

Existing rendezvous approaches are typically based on trajectory tracking or optimal control formulations. Classical Proportional–Integral–Derivative (PID) and Linear Quadratic Regulator (LQR) controllers provide satisfactory performance under linearized dynamics but do not explicitly enforce terminal constraints and exhibit sensitivity to parameter tuning in nonlinear regimes (Wei et al. (2025); Proia et al. (2025)). Model Predictive Control (MPC) is widely adopted due to its ability to handle multi-constraint optimization in real-time, and has been successfully applied to docking and target-tracking problems in both aerial and space scenarios (Li et al. (2017); Quartullo et al. (2024)). However, MPC formulations generally rely on fixed or predefined prediction horizons, which prevent explicit online optimization of the maneuver duration and may limit convergence accuracy and efficiency in dynamic rendezvous tasks. Direct transcription techniques provide accurate discretizations of the optimal control problem but introduce approximation errors related to mesh discretization, often leading to discontinuities in the control profile and reduced terminal precision (Luo et al. (2014); Proia et al. (2023)).

Optimal control formulations with free-final-time allow the maneuver duration to be treated as a decision variable and determined within the optimization process (Garcia et al. (2015)). Several works have investigated time-optimal (Verschueren et al. (2017)) and variable-horizon strategies (Quartullo et al. (2024)) for rendezvous problems. However, existing approaches typically rely on discrete-time implementations or structured predictive schemes, where the prediction horizon is fixed, heuristically adjusted, or only partially optimized online. As a result, the integration of free-final-time formulations within receding horizon control schemes remains challenging in the presence of nonlinear dynamics, input constraints, and time-varying terminal conditions. In particular, the joint online optimization of trajectory and completion time within a continuous-time receding horizon framework remains largely unexplored for autonomous rendezvous in HDI scenarios, where the terminal configuration evolves with human motion.

To address this gap, we propose a continuous-time receding horizon control for safe and ergonomic human-drone rendezvous with free-final-time. The terminal time is treated as an optimization variable and updated online at each replanning step, enabling simultaneous optimization of trajectory and maneuver duration under time-varying interaction conditions. Compared with fixed-horizon predictive control formulations, the maneuver duration emerges directly from the optimization process and adapts to the system state and constraints, improving convergence accuracy, energy efficiency, and overall performance.

The main contributions are summarized as follows:

(i) design of a continuous-time receding horizon control framework with free-final-time for autonomous rendezvous;
(ii) integration of safety, ergonomic feasibility, and energy efficiency within a unified optimal control framework for HDI in indoor industrial environments.

2. SYSTEM MODELING

This section details the drone dynamics (Section 2.1) and the human representation (Section 2.2). Among different UAVs, the drone considered in this work is a quadrotor, representative of the class of multirotor aerial vehicles commonly employed in indoor environments due to their maneuverability and compact design.

2.1 Quadrotor Dynamics

A quadrotor, also referred to as a quadcopter, consists of four rotors arranged in two counter-rotating pairs at the corners of a rigid square frame. In the context of HDI, the inner attitude control loop ensures near-hover stability, enabling a clear separation between the slow translational dynamics and the fast rotational one. Accordingly, the following formulation focuses on the translational dynamics of the drone, modeled as a three–degree–of–freedom system evolving along the Cartesian axes of the East–North–Up (ENU) inertial reference frame, while the orientation control is assumed to maintain nominal hover conditions.

The quadrotor is modeled as a symmetric rigid body under small-angle assumptions, neglecting aerodynamic drag, propeller–airframe coupling, and ground effects, as outlined in Sabatino (2015) and Das et al. (2009).

The state vector of the drone is defined as

$$s^{\mathrm{d}}(t) = \left(p^{\mathrm{d}}(t)^\top \; \dot{p}^{\mathrm{d}}(t)^\top\right)^\top \in \mathbb{R}^6, \tag{1}$$

where $p^{\mathrm{d}}(t) = \left(x^{\mathrm{d}}(t), y^{\mathrm{d}}(t), z^{\mathrm{d}}(t)\right)^\top$ denotes the position in the inertial frame, and $\dot{p}^{\mathrm{d}}(t) = \left(\dot{x}^{\mathrm{d}}(t), \dot{y}^{\mathrm{d}}(t), \dot{z}^{\mathrm{d}}(t)\right)^\top$ its linear velocity, compactly indicated as $v^{\mathrm{d}}(t) \coloneqq \dot{p}^{\mathrm{d}}(t)$.

The control input is expressed as

$$u^{\mathrm{d}}(t) = \left(a^{\mathrm{d}}_x(t) \; a^{\mathrm{d}}_y(t) \; a^{\mathrm{d}}_z(t)\right)^\top \in \mathbb{R}^3, \tag{2}$$

representing the equivalent translational accelerations along the body axes, generated by the attitude control layer to regulate the drone motion in the inertial frame.

Employing the state vector $s^{\mathrm{d}}(t)$ and the control vector $u^{\mathrm{d}}(t)$, the translational dynamics is expressed as

$$\begin{aligned} \dot{s}^{\mathrm{d}}(t) &= f^{\mathrm{d}}\big(s^{\mathrm{d}}(t), u^{\mathrm{d}}(t), t\big) \\ &= \begin{bmatrix} \dot{p}^{\mathrm{d}}(t) \\ -\dfrac{\mathrm{k}^{\mathrm{d}}}{\mathrm{m}^{\mathrm{d}}}\, \dot{p}^{\mathrm{d}}(t) + \dfrac{1}{\mathrm{m}^{\mathrm{d}}}\, u^{\mathrm{d}}(t) \end{bmatrix} \\ s^{\mathrm{d}}(t_0) &= \mathrm{s}^{\mathrm{d},0}. \end{aligned} \tag{3}$$

where m^{d} is the drone mass and $\mathrm{k}^{\mathrm{d}} > 0$ is a viscous damping coefficient accounting for aerodynamic resistance. The function $f^{\mathrm{d}} : \mathbb{R}^{10} \to \mathbb{R}^6$ represents the nonlinear, time-varying translational dynamics of the quadrotor, and $\mathrm{s}^{\mathrm{d},0} \in \mathbb{R}^6$ denotes the initial state of the drone at the initial time t_0.

The explicit time dependence in (3) allows the model to capture nonstationary operating conditions, such as adaptive speed modulation or time-varying environmental effects arising in HDI scenarios.

2.2 Human Representation

The human operator moves along a smooth, time–parametrized trajectory that serves as an external input to the rendezvous problem. We assume that the operator's position evolves as $\mathrm{p}^{\mathrm{h}}(t) = (\mathrm{x}^{\mathrm{h}}(t), \mathrm{y}^{\mathrm{h}}(t))^\top \in \mathbb{R}^2$, whose time derivative $\mathrm{v}^{\mathrm{h}}(t) \coloneqq \dot{\mathrm{p}}^{\mathrm{h}}(t) = (\dot{\mathrm{x}}^{\mathrm{h}}(t), \dot{\mathrm{y}}^{\mathrm{h}}(t))^\top$ represents the instantaneous velocity. The operator's heading (i.e., the orientation of the body with respect to the global frame) $\psi^{\mathrm{h}}(t)$ is assumed to remain aligned with the instantaneous direction of motion, meaning that the operator faces and moves forward along the trajectory without backward or lateral motion (i.e., $\psi^{\mathrm{h}}(t) = \mathrm{atan2}(\dot{\mathrm{y}}^{\mathrm{h}}(t), \dot{\mathrm{x}}^{\mathrm{h}}(t))$). The explicit modeling of human biomechanics, motion prediction, and behavioral variability is considered beyond the scope of our work.

3. RENDEZVOUS PROBLEM FORMULATION

The rendezvous trajectory planning problem for HDI is formulated in continuous time as a free-final-time optimal control problem. The objective is to minimize actuation energy and translational kinetic effort while ensuring proximity safety and ergonomic feasibility of the final interaction configuration.

Let $[t_0, t_f]$ denote the finite time horizon, where $t_0 \in \mathbb{R}$ is the initial time and $t_f \in \mathbb{R}$ is the terminal time. The horizon is finite, but the terminal time t_f is a free variable to be determined as part of the optimization process. The continuous time variable is denoted by $t \in [t_0, t_f]$.

The overall formulation, presented in the following, integrates three essential components: (i) a safety regulation law ensuring distance–based speed modulation (Section 3.1); (ii) an ergonomic constraint defining the admissible terminal configuration (Section 3.2); and (iii) energy and motion efficiency terms promoting smooth and energy–aware trajectories (Section 3.3).

All components are integrated within the optimal control problem presented in Section 3.4, combining safety regulation, ergonomic feasibility, and energy efficiency.

3.1 Safety Regulation

To ensure safe HDI, the drone motion must comply with a minimum protective distance from the operator and adapt its speed to the instantaneous separation distance. We define a continuous safety regulation law that prescribes a time-varying safe speed profile $v_{\text{safe}}(t)$, modeled as a nonincreasing function of the relative positions of the two agents:

$$v_{\text{safe}}(t) = \tilde{v}_{\text{safe}}\big(p^{\text{d}}(t), \mathrm{p}^{\text{h}}(t)\big) = \begin{cases} 0, & d(t) \le \mathrm{d}_{\text{stop}}, \\ \mathrm{v}_{\text{cruise}} \dfrac{d(t) - \mathrm{d}_{\text{stop}}}{\mathrm{d}_{\text{cruise}} - \mathrm{d}_{\text{stop}}}, & \mathrm{d}_{\text{stop}} < d(t) < \mathrm{d}_{\text{cruise}}, \\ \mathrm{v}_{\text{cruise}}, & d(t) \ge \mathrm{d}_{\text{cruise}}, \end{cases} \tag{4}$$

where the instantaneous human–drone separation is defined as

$$d(t) = \|p^{\text{d}}(t) - \mathrm{p}^{\text{h}}(t)\|, \tag{5}$$

and the parameters d_{stop} and $\mathrm{d}_{\text{cruise}}$ denote respectively the inner and outer bounds of the protective region, while $\mathrm{v}_{\text{cruise}} > 0$ represents the nominal cruise speed. The profile ensures smooth and monotonic deceleration as the drone approaches the operator, avoiding abrupt speed changes and maintaining dynamic feasibility. Safety regulation is integrated into the cost function through a quadratic penalty term based on $v_{\text{safe}}(t)$, enabling the optimization process to continuously adapt the drone speed while ensuring compliance with proximity requirements.

3.2 Ergonomic Feasibility

Ergonomic feasibility defines the spatial configurations that allow the drone to approach the human operator under admissible postural effort. The formulation extends the ergonomic model proposed by Proia et al. (2025), which employs the RULA index to quantify the postural condition as a function of the human–drone relative pose. An ergonomic volume $\mathcal{V}^\star \subseteq \mathbb{R}^3$ is defined within the collaborative work zone, representing the set of spatial configurations corresponding to the minimum RULA index that can be comfortably reached by the operator.

The volume is expressed in the human reference frame $\{F^{\text{h}}(t)\}$, rigidly attached to the body with origin at the position $\mathrm{p}^{\text{h}}(t)$, located between the operator's feet. The axes are aligned with the anthropometric directions, with x oriented forward along the torso, y directed toward the left arm, and z oriented vertically upward. In the $\{F^{\text{h}}(t)\}$ coordinates, the ergonomic volume $\mathcal{V}^\star$ lies in front of the torso and is slightly shifted toward the operator's dominant-hand side, corresponding to the natural reaching area for frontal manipulation.

The relative configuration between the drone and the operator within $\{F^{\text{h}}(t)\}$ is described by the ergonomic displacement vector

$$\delta = \big(\delta_x, \delta_y, \delta_z\big)^\top \in \mathcal{V}^\star \subset \mathbb{R}^3, \tag{6}$$

where δ_x is the forward offset relative to the torso, δ_y is the lateral offset toward the operator's dominant-hand side, and δ_z is the vertical offset relative to the human frame origin.

Since the rendezvous control problem is formulated in the ENU inertial reference frame, a coordinate transformation maps the ergonomically admissible region into the same frame:

$$p^{\text{d}}(t_f) = \mathrm{p}^{\text{h}}(t_f) + R^{\text{h}}(t_f)\,\delta, \qquad \delta \in \mathcal{V}^\star, \tag{7}$$

where $R^{\text{h}}(t_f) \in SO(3)$ denotes the rotation matrix describing the orientation of $\{F^{\text{h}}(t)\}$ with respect to the ENU frame, given by

$$R^{\text{h}}(t_f) = \begin{bmatrix} \cos\psi^{\text{h}}(t_f) & -\sin\psi^{\text{h}}(t_f) & 0 \\ \sin\psi^{\text{h}}(t_f) & \cos\psi^{\text{h}}(t_f) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{8}$$

The combined effect of translation and rotation preserves the geometry of the ergonomic volume in the human frame and ensures a consistent positioning in the ENU frame as the operator moves.

Accordingly, the terminal configuration ensuring ergonomic admissibility is defined by (7), which constrains the drone's final position to lie within the ergonomically admissible region $\mathcal{V}^\star$. A rest-to-hover condition is further imposed as

$$v^{\text{d}}(t_f) = 0. \tag{9}$$

3.3 Energy and Motion Efficiency

Energy consumption represents a key aspect of HDI, as it directly influences operational efficiency and mechanical stress on the aerial platform. In the proposed framework, energy efficiency is achieved by penalizing the control effort, while additional kinematic terms promote smooth and dynamically feasible motion, as detailed in Section 3.4.

3.4 Optimal Control Problem Definition

The rendezvous optimal control formulation integrates safety regulation and energy–motion efficiency within the cost function, while enforcing ergonomic admissibility through terminal constraints.

The performance index proposed in this work is defined as

$$J^{\text{d}}\big(u^{\text{d}}(\cdot), t_f, \delta\big) = \int_{t_0}^{t_f} \Big(\|u^{\text{d}}(t)\|^2_{\mathrm{R}} + \|v^{\text{d}}(t)\|^2_{\mathrm{Q}} + \|v^{\text{d}}(t) - v_{\text{safe}}(t)\hat{v}^{\text{d}}(t)\|^2_{\mathrm{S}} \Big)\, dt. \tag{10}$$

where $\mathrm{R} \succ 0$, $\mathrm{Q} \succeq 0$, and $\mathrm{S} \succeq 0$ are weighting matrices. The first term in the integral penalizes the control effort, the second promotes kinematic smoothness, and the third enforces compliance with the safe speed profile $v_{\mathrm{safe}}(t)$. The unit vector $\hat{v}^{\mathrm{d}}(t) = v^{\mathrm{d}}(t)/\|v^{\mathrm{d}}(t)\|$ represents the instantaneous heading direction of the drone.

The feasible set of trajectories is determined by the translational dynamics introduced in (3), the safety regulation law (4), and the ergonomic terminal conditions given in (7)-(9). Control inputs are bounded within the admissible actuation set

$$u^{\mathrm{d}}(t) \in \mathcal{U} \subset \mathbb{R}^3, \qquad t \in [t_0, t_f], \tag{11}$$

where $\mathcal{U}$ denotes the set of admissible control functions $u^{\mathrm{d}} : [t_0, t_f] \to \mathbb{R}^3$ accounting for thrust and attitude limitations of the aerial platform.

The complete human–drone rendezvous trajectory planning problem is then formulated as

$$\begin{aligned} \min_{u^{\mathrm{d}}(\cdot)\in\mathcal{U},\, t_f \ge t_0,\, \delta\in\mathcal{V}^\star} \quad & J^{\mathrm{d}}\big(u^{\mathrm{d}}(\cdot), t_f, \delta\big) \\ & \text{subject to} \\ & \text{constraints (3), (7), (9), (11).} \end{aligned} \tag{12}$$

4. RENDEZVOUS PROBLEM REFORMULATION

Problem (12) is a nonconvex, free–final–time optimal control problem with bounded inputs and terminal constraints. The displacement vector δ introduces an implicit coupling between control variables and the terminal state of the drone, preventing a direct solution of the original formulation.

A parametric reformulation is introduced by treating the ergonomic displacement vector $\delta \in \mathcal{V}^\star$ as a fixed parameter. The resulting optimization problem is expressed as

$$\min_{\delta\in\mathcal{V}^\star} \widetilde{J}^{\mathrm{d}}(\delta), \tag{13}$$

where the value function $\widetilde{J}^{\mathrm{d}}(\delta)$ is defined as the optimal cost of a free–final–time optimal control problem:

$$\begin{aligned} \widetilde{J}^{\mathrm{d}}(\delta) = \min_{u^{\mathrm{d}}(\cdot)\in\mathcal{U},\, t_f \ge t_0} \quad & J^{\mathrm{d}}\big(u^{\mathrm{d}}(\cdot), t_f, \delta\big) \\ & \text{subject to} \\ & \text{constraints (3), (7), (9), (11).} \end{aligned} \tag{14}$$

Function $\widetilde{J}^{\mathrm{d}}(\delta)$ represents the optimal value of the performance index in (10), combining control effort, motion smoothness, and safety requirements for a given ergonomic configuration δ.

Due to nonlinear dynamics and the coupling among control input, state trajectory, and free terminal time, an exact evaluation of $\widetilde{J}^{\mathrm{d}}(\delta)$ does not admit a closed-form expression and requires numerical solution of the associated optimal control problem. Available approaches include indirect methods based on the Pontryagin Maximum Principle (Kopp (1962)) and direct formulations, such as the time-optimal rendezvous (Quartullo et al. (2024)).

5. RENDEZVOUS REPLANNING APPROACH

In industrial warehouses, human motion usually follows fixed and structured paths, such as aisles or corridors connecting storage zones and palletizing areas. The corresponding trajectories can be predefined and known in advance, allowing the drone to plan the rendezvous within a spatially constrained environment. Nevertheless, the operator's walking speed may vary according to task execution, carried load, and environmental conditions, resulting in a nonuniform progression of the human position along the predefined path.

Algorithm 1 Free-Final-Time Receding Horizon Rendezvous Replanning

Input: t_0, $\mathrm{s}^{\mathrm{d},0}$, and $\mathrm{p}^{\mathrm{h}}(t), \psi^{\mathrm{h}}(t), t \ge t_0$
Parameters: ε, Δt
Output: $u^{\mathrm{d},\star}(t)$

1: $t_j = t_0$
2: **while** $\|p^{\mathrm{d}}(t_j) - \mathrm{p}^{\mathrm{h}}(t_j)\| > \varepsilon$ **do**
3: $\quad t_0 = t_j$
4: $\quad$ Evaluate $s^{\mathrm{d}}(t_j)$
5: $\quad$ Update $\mathrm{p}^{\mathrm{h}}(t), \psi^{\mathrm{h}}(t), t \ge t_0$
6: $\quad$ Solve (13)
7: $\quad$ Apply $u^{\mathrm{d},\star}(t), t \in [t_j,\, t_j + \Delta t)$
8: $\quad t_j \leftarrow t_j + \Delta t$
9: **end while**

To preserve safety, ergonomics compliance, and energy efficiency as the human pose evolves, the rendezvous optimal control problem (13) is iteratively solved within a receding–horizon scheme. At each replanning cycle, the problem is integrated from the current time instant t_j over the optimal prediction horizon T_j, which is determined as part of the solution.

Algorithm 1 – *Free–Final–Time Receding Horizon Rendezvous Replanning* iteratively updates the optimal drone trajectory in response to the evolving human motion. The algorithm takes as input the drone initial state $s^{\mathrm{d}}(t_0)$, together with the human position $\mathrm{p}^{\mathrm{h}}(t)$ and heading $\psi^{\mathrm{h}}(t)$, with $t \ge t_0$. At each replanning cycle, the current drone state $s^{\mathrm{d}}(t_j)$ and human motion variables $\mathrm{p}^{\mathrm{h}}(t), \psi^{\mathrm{h}}(t), t \ge t_0$ are evaluated (Algorithm 1, lines 4-5). The rendezvous optimal control problem (13) is then solved (Algorithm 1, line 6), providing the optimal control input $u_j^{\mathrm{d},\star}(t)$. The control is applied over the interval $[t_j,\, t_j + \Delta t)$ (Algorithm 1, line 7), after which the next replanning instant is updated as $t_j = t_j + \Delta t$ (Algorithm 1, line 8). The process is repeated until the rendezvous condition $\|p^{\mathrm{d}}(t_j) - \mathrm{p}^{\mathrm{h}}(t_j)\| \le \varepsilon$ is satisfied, where ε denotes the distance tolerance defining the terminal proximity between the drone and the human.

6. NUMERICAL EXPERIMENTS

6.1 Setup

The simulations aim at validating the performance of the proposed human–drone rendezvous control approach in a warehouse environment. The considered scenario reproduces a collaborative pick-and-delivery operation, where drones assist human operators during palletizing and material-handling tasks. All simulations are executed in MATLAB R2024b, where the control framework integrates the nonlinear optimizer `fmincon` for the parametric optimization with the collocation-based solver `bvp4c` for the numerical solution of problem (13).

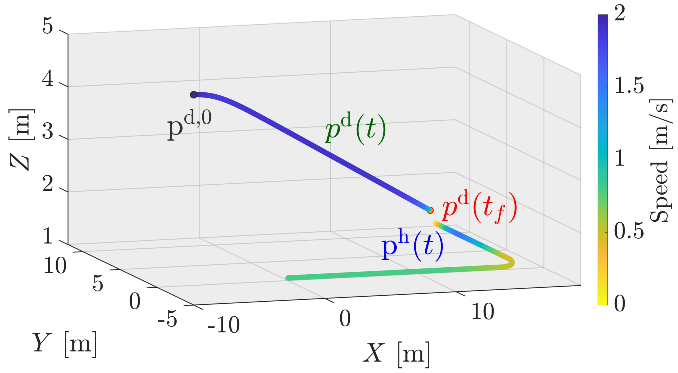

Fig. 1. Drone and human trajectories during the rendezvous maneuver with colormap based on speed.

The quadrotor operates within the ENU frame centered at the lower-left vertex of the warehouse workspace. It is modeled as a translational point mass with viscous damping coefficient $\mathrm{k}^{\mathrm{d}} = 0.05$ and total mass $\mathrm{m}^{\mathrm{d}} = 1.00$ kg. The initial position is set to $\mathrm{p}^{\mathrm{d},0} = (-10.00, -5.00, 5.00)^\top$ m and the initial velocity to $\mathrm{v}^{\mathrm{d},0} = (2.00, 0.00, 0.00)^\top$ m/s, corresponding to a forward motion along the x direction. The admissible control set is defined as $\mathcal{U} = \{u^{\mathrm{d}} \in \mathbb{R}^3 \mid \|u^{\mathrm{d}}\|_\infty \leq 3 \text{ m/s}^2\}$, which reflects the drone's actuation limits.

The human operator is a right-handed adult male at the 50th percentile of height and follows a time–parametrized trajectory $\mathrm{p}^{\mathrm{h}}(t)$ composed of an initial rectilinear segment along the X–axis, a smooth circular transition, and a subsequent translation along the Y–axis, representative of a collaborative pick-and-delivery task in an industrial warehouse. The walking speed $\mathrm{v}^{\mathrm{h}}(t)$ varies between 0.8 m/s and 1.5 m/s, consistently with typical ranges observed in human locomotion during collaborative operations.

The ergonomic volume $\mathcal{V}^\star$ is defined in the human reference frame $\{F^{\mathrm{h}}(t)\}$ and bounded within the ranges $[-0.20, 0.10]$ m, $[0.35, 0.50]$ m, and $[1.25, 1.35]$ m along the forward, lateral, and vertical directions, as determined through the RULA ergonomic analysis in Proia et al. (2023, 2025). Safety constraints are implemented by means of a distance-dependent speed limit $v_{\mathrm{safe}}(t)$, characterized by $\mathrm{d}_{\mathrm{stop}} = 0.10$ m, $\mathrm{d}_{\mathrm{cruise}} = 3.00$ m, and $\mathrm{v}_{\mathrm{cruise}} = 2.00$ m/s.

6.2 Results

The following results illustrate the overall behavior of the proposed control framework, highlighting the capability of the system to adapt to human motion while preserving safety, ergonomics, and efficiency requirements.

The rendezvous operation shows that the controller continuously replans at each $\Delta t = 1$ s to account for the nonuniform human motion, characterized by speed variations along the path. The resulting trajectory remains smooth and dynamically feasible, ensuring consistent convergence despite the time-varying terminal conditions.

Safety compliance is achieved through the continuous adaptation of the drone speed to the human–drone relative distance, as prescribed by the distance-dependent regulation law. As shown in Fig. 1, the color variation along the trajectory reflects the speed modulation, with the drone speed progressively decreasing as the separation distance reduces, ensuring smooth deceleration during the final approach phase.

Ergonomic feasibility is enforced through the terminal constraint defining the admissible interaction region. The final drone position lies within the ergonomic volume $\mathcal{V}^\star$, corresponding to a comfortable reach configuration identified through the RULA-based assessment. Convergence to an ergonomically admissible configuration is achieved at the end of the maneuver, with

$$p^{\mathrm{d}}(t_f) = (-0.07 \;\; 0.43 \;\; 1.31)^\top \text{ m}.$$

The corresponding final position error is equal to 0.44 m, indicating high terminal precision under time-varying conditions.

Overall, the proposed approach generates smooth and energy-efficient trajectories, achieving accurate terminal convergence while maintaining safety and ergonomic compliance under time-varying interaction conditions.

7. CONCLUSION

A continuous-time receding horizon control framework with free-final-time has been presented for autonomous human–drone rendezvous in industrial Human–Drone Interaction. The proposed approach integrates safety, ergonomic, and energy-efficiency objectives within a unified optimal control formulation, while enabling real-time adaptation through online optimization of both trajectory and maneuver duration. Simulation results in a warehouse scenario demonstrate that the controller ensures safe and ergonomic interaction with a moving operator, generating smooth and energy-efficient trajectories with accurate terminal convergence.

Future work will address predictive modeling of operator motion, adaptive replanning mechanisms, and improved robustness under more complex and uncertain interaction conditions.

REFERENCES

Das, A., Subbarao, K., and Lewis, F. (2009). Dynamic inversion with zero-dynamics stabilisation for quadrotor control. *IET control theory & applications*, 3(3), 303–314.

Garcia, G., Keshmiri, S., and Huang, W. (2015). Recovery of an aircraft from the loss of control using open final time dynamic optimization and receding horizon control. In *AIAA Guidance, Navigation, and Control Conference*, 1545.

Kopp, R.E. (1962). Pontryagin maximum principle. In *Mathematics in Science and Engineering*, volume 5, 255–279. Elsevier.

Li, Q., Yuan, J., Zhang, B., and Gao, C. (2017). Model predictive control for autonomous rendezvous and docking with a tumbling target. *Aerospace Science and Technology*, 69, 700–711.

Lucci, N., Lacevic, B., Zanchettin, A.M., and Rocco, P. (2020). Combining speed and separation monitoring with power and force limiting for safe collaborative robotics applications. *IEEE Robotics and Automation Letters*, 5(4), 6121–6128.

Luo, Y., Zhang, J., and Tang, G. (2014). Survey of orbital dynamics and control of space rendezvous. *Chinese Journal of Aeronautics*, 27(1), 1–11.

McAtamney, L. and Corlett, E.N. (1993). Rula: a survey method for the investigation of work-related upper limb disorders. *Applied Ergonomics*, 24(2), 91–99.

Mirri, S., Prandi, C., and Salomoni, P. (2019). Human-drone interaction: state of the art, open issues and challenges. In *Proceedings of the ACM SIGCOMM 2019 Workshop on Mobile AirGround Edge Computing, Systems, Networks, and Applications*, 43–48.

Proia, S., Carli, R., Cavone, G., and Dotoli, M. (2021). Control techniques for safe, ergonomic, and efficient human-robot collaboration in the digital industry: A survey. *IEEE Transactions on Automation Science and Engineering*, 19(3), 1798–1819.

Proia, S., Cavone, G., Scarabaggio, P., Carli, R., and Dotoli, M. (2023). Safety compliant, ergonomic and time-optimal trajectory planning for collaborative robotics. *IEEE Transactions on Automation Science and Engineering*.

Proia, S., Cavone, G., Scarabaggio, P., Carli, R., and Dotoli, M. (2025). An integrated control framework for safe and ergonomic human-drone interaction in industrial warehouses. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.

Quartullo, R., Bianchini, G., Garulli, A., and Giannitrapani, A. (2024). Robust time-optimal model predictive control for rendezvous with a moving target. In *Conf. on Decis. and Control*, 7510–7515. IEEE.

Sabatino, F. (2015). Quadrotor control: modeling, nonlinearcontrol design, and simulation.

Verschueren, R., Ferreau, H.J., Zanarini, A., Mercangöz, M., and Diehl, M. (2017). A stabilizing nonlinear model predictive control scheme for time-optimal point-to-point motions. In *2017 IEEE 56th annual conference on decision and control (CDC)*, 2525–2530. IEEE.

Wawrla, L., Maghazei, O., and Netland, T. (2019). Applications of drones in warehouse operations. *Whitepaper. ETH Zurich, D-MTEC.*

Wei, Z., Yang, J., Wen, H., Jin, D., and Chen, T. (2025). Nonlinear optimal control for spacecraft rendezvous and docking using symplectic numerical method. In *Actuators*, volume 14, 75. MDPI AG.